

GENERAL THETA FOUNDATIONS I: ORBIT GEOMETRY AND ENTROPY BUDGETS FOR FINITE MEMORY

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ABSTRACT. We prove a finite-memory occupation theorem for orthogonal representations of compact groups. A norm gap away from invariant functions and a uniform orbital small-ball bound of exponent s force the gap-weighted number of k -state epochs to be $O(k^{2/s})$. The competing transducer may have arbitrary hidden states and unrelated stochastic rows at every epoch. The small-ball hypothesis is required on the directions of all possible conditional means, not only on the original orbit. For conjugation on traceless Hermitian q by q matrices we establish it uniformly over all spectra with exponent $2(q-1)$. A projective-net construction then gives matching $\Theta(N^{q-1})$ classical labels for finite algebraic unitary alphabets with a full group spectral gap, at exact output and at every fixed sufficiently small wordwise error. The same numerical behavior has a fixed q -dimensional quantum realization. We also give a separate six-command rational three-dimensional example with an analytic numerical gap certificate from the classical Ramanujan bound, yielding an explicit finite-horizon lower constant. The resource is nonuniform clocked atomic-row width, not uniform computational space. Gaussian entropy identities, classical expansion and coding antecedents are distinguished from the orbital occupation and synthesis conclusions.

1. INTRODUCTION

A finite probabilistic device can preserve a linear statistic through a long sequence of transformations without retaining the underlying point. Consequently the dimension of the statistic does not determine the size of an exact positive realization. The distinction is particularly visible when the transformation group expands functions on its orbits. Mixing creates entropy, while replacement by finitely many conditional means removes it. An occupation inequality can compare these operations without assigning an observable vector to each hidden basis state.

The spherical form of this argument gives the order $N^{(d-1)/2}$ under a full spherical spectral gap [20]. Two issues arise beyond the sphere. First, invariant densities need not be radial. Second, a conditional mean of orbit points need not be another orbit point. The second issue is decisive for unitary conjugation: a conditional mean of pure states usually has several nontrivial eigenvalues. Applying a pure-state cap estimate directly to these means would be invalid.

We resolve both issues through orbital averaging on the whole ambient representation and a spectral-projector estimate uniform over all traceless Hermitian directions. The resulting exponent is intrinsic to the smallest nontrivial conjugation orbit, rather than the dimension of the ambient sphere. A construction on complex projective space matches this lower exponent. Thus the theorem is not obtained by substituting $q^2 - 1$ into a spherical formula.

1.1. Principal statements. Let G act orthogonally on $V = \mathbb{R}^D$. Write Π_G for Haar averaging and suppose that a finitely supported command law has squared norm gap $g > 0$ away from Π_G

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on $L^2(V)$. Assume, for $s > 0$ and $A \geq 1$,

$$(1) \quad \sup_{u,v \in \mathbb{S}(V)} m_G\{a : \|au - v\| \leq r\} \leq Ar^s \quad (0 < r \leq 1).$$

The input word is universally quantified; the command law is chosen only for the lower proof. Theorem 3.5 proves

$$(2) \quad \sum_{t: K_t \leq k} g_t \leq 2048D(16A)^{2/s} \kappa^{-3-2/s} k^{2/s},$$

whenever a terminal correlation forces $\mathbb{E}\|\mathbb{E}[Y_N \mid S_N]\| \geq \kappa$. Widths at the other epochs are unrestricted. The gap may vary with time.

For $q \geq 2$, put $D = q^2 - 1$ and consider the normalized pure-state orbit in $\text{Herm}_0(q)$, equipped with its Hilbert–Schmidt inner product. A fixed finite unitary alphabet drives this orbit. One final query asks for a coordinate of the resulting vector, with fixed amplitude $0 < \rho \leq 1/10$.

Theorem 1.1 (Projective-orbit memory order). *Suppose the command alphabet supports a probability measure with a norm gap on $L_0^2(\text{SU}(q))$. For every fixed $0 \leq \varepsilon < \rho/(2\sqrt{q^2 - 1})$, the least classical width satisfies*

$$(3) \quad W_{N,\varepsilon} = \Theta_{q,\rho,\varepsilon,\mu}(N^{q-1}).$$

The lower bound permits arbitrary hidden states, time-dependent rows and horizon-dependent design. The exact upper construction uses common command rows and an N -dependent final decoder. There are explicit finite algebraic alphabets satisfying the hypothesis for every q . The behavior has a fixed quantum realization of dimension q ; its classical label bits are $(q - 1) \log_2 N + O(1)$.

The proof, including the alphabet and the fixed quantum device, is in Section 5. For $q > 2$ the exponent $q - 1$ differs from the ambient-sphere exponent $(q^2 - 2)/2$. The group-gap hypothesis here is qualitative; its existence follows from an external expansion theorem. It is not inferred merely from density of the generated subgroup.

A second application addresses effective constants. Section 6 uses the classical six V -gates, whose adjoint rotations have rational entries. The Lubotzky–Phillips–Sarnak bound gives a certified squared gap $4/9$ on the sphere. For amplitude $1/10$ the explicit consequence is

$$(4) \quad W_{N,\varepsilon} \geq \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N.$$

This is a different alphabet from the earlier five-gate example, whose numerical gap is not evaluated here. The arithmetic spectral estimate is an external theorem, not a numerical truncation of spherical harmonics.

1.2. Conventions and attribution. All growth statements concern available persistent labels. The external epoch, the horizon, read-only row tables, their construction, arithmetic and exact atomic sampling are not charged. One query is selected after the command word; no independent joint answers are required. A finite random-bit implementation is a different resource problem.

Ingredient	Role in the argument
Gaussian score and conditional means	Classical identities; the one-sided quadratic entropy consequence is reproduced, not used as a priority claim.
Group spectral gaps	External inputs, with a separate numerical Ramanujan certificate for the six-gate example.
Orbital occupation	The representation-level extension, using uniform small balls for all centroid directions.
Hermitian conjugation	A uniform all-spectra estimate and matching projective synthesis give the $q - 1$ width exponent.
Quantum operations	Standard reset, conjugation and two-outcome measurement; the width lower bound concerns the classical competitor.

The comparison with causal coding, Gaussian conditional-expectation entropy, state aggregation and probabilistic branching programs is made at the level of the optimized objects in Section 7. **Notation.** N is the command horizon; K_t is the available label count; W is its maximum. D is the real ambient dimension and s is the orbital small-ball exponent. q denotes quantum dimension, while q_t below denotes a conditional-centroid norm. Entropy and log use natural units; $\log_2$ is reserved for label bits.

2. THE EXPERIMENT AND ITS POSITIVE REALIZATION

Let G be a compact group, let $U : G \rightarrow O(V)$ be a continuous orthogonal representation on $V = \mathbb{R}^D$, and fix a finite command alphabet $\mathcal{A} \subset G$. A finite seed set $\mathcal{X} \subset \mathbb{S}(V)$ is given initially. For $w = (a_1, \dots, a_N)$ write $U_w = U_{a_N} \cdots U_{a_1}$. Relative to an orthonormal basis $e_1, \dots, e_D$, the prescribed numerical experiment is

$$(5) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\}, \quad j \in [D], \quad 0 < \rho \leq 1/10.$$

All seeds, words and queries are required, not only almost-sure words under one source law. The probabilities are between $(1 - \rho)/2$ and $(1 + \rho)/2$.

Definition 2.1 (Clocked atomic-row realization). A horizon- N machine consists of finite sets $\mathcal{S}_t$ with $|\mathcal{S}_t| = K_t$, a stochastic initialization E_x on $\mathcal{S}_0$, stochastic matrices $T_{t,a} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1}$, and columns $d_j : \mathcal{S}_N \rightarrow [-1, 1]$. Its mean response is

$$(6) \quad E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, U_w x \rangle$ by at most 2ε for every (x, w, j) . Available labels, including padding, count. All rows, including unreachable ones, are normalized. Only the updated label persists. The seed, word prefix, old random choices and any data-dependent protocol information cannot be consulted again unless represented by that label. The external time, horizon, fixed program and exact sampling from its rows are free. The final binary answer may be sampled when j arrives; if retained, it uses a two-label final cut. Write $W_{N,\varepsilon}$ for the minimum of $\max(2, K_0, \dots, K_N)$.

The state alphabets and matrices may be chosen anew for each N , but the command symbols and their experimental meanings are fixed. Nothing requires one autonomous device or one decoder to serve all stopping lengths. No adaptive choice of the command is optimized in this model. A complex unitary representation can be treated by its realification; all inner products in the entropy argument are real.

Fix one seed x and an independent command law at each epoch for the converse. Let $Y_t = U_{a_t} \cdots U_{a_1} x$ and let S_t be the actual machine state under this test law. Define

$$(7) \quad Z_t = \mathbb{E}[Y_t \mid S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2.$$

These random vectors and their laws are analytical devices, not an additional runtime register or oracle.

Lemma 2.2 (Quadratic budget and terminal calibration). *For the fresh command a , put $X_t = U_a Z_t$. Then*

$$(8) \quad Z_{t+1} = \mathbb{E}[X_t \mid S_{t+1}], \quad \Delta_t := \mathbb{E}\|X_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The sequences q_t, v_t are nonincreasing and start at one. A machine with error at most ε satisfies

$$(9) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon \quad (0 \leq t \leq N).$$

Proof. The current command is independent of the past, and the new label is generated from (S_t, a) by a common row. Conditioning first on (S_t, a, S_{t+1}) and then on S_{t+1} proves the first identity. The orthogonal projection identity for conditional expectation gives the second; Jensen and orthogonality give both monotonicities. The fixed seed has unit norm, even with randomized initialization. For calibration form $d = (d_1, \dots, d_D)$ only in the proof, so $\|d\| \leq \sqrt{D}$. Conditioning on the complete word and using (5)–(6) gives $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\sqrt{D}\varepsilon$. On conditioning instead on S_N , this is at most $\sqrt{D}q_N$. This proves (9). It does not require a joint measurement of the D coordinates. $\square$

For each fixed profile, the optimum actual error is attained: all stochastic rows and terminal columns form a finite product of compact sets and the objective is a maximum of finitely many continuous functions. Thus the subsequent lower bounds concern the actual optimization, not a sum of local error certificates.

3. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(10) \quad \mathcal{U}_a f(z) = f(U_a^{-1}z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1}z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(11) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma 3.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (11) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1}z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1}z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (11). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

3.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma 3.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(12) \quad \mathbf{h}(X + \tau G_D) - \mathbf{h}(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(13) \quad \nabla^2 \log f(y) = -\tau^{-2}I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2}I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2}I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $\mathbf{h}(g) \leq -\int g \log f = \mathbb{E}\phi(X)$, whereas $\mathbb{E}\phi(M) = \mathbf{h}(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (12). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 14]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma 3.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(14) \quad \mathbf{h}(P_t f) - \mathbf{h}(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $\mathbf{h}(h) - \mathbf{h}(f) = \sum_a p_t(a) D(f_a \parallel h)$. For probability densities u, v , $D(u \parallel v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (11) and $\|\sqrt{f}\|_2 = 1$. $\square$

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

3.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(15) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma 3.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(16) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (15) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi\sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. $\square$

Theorem 3.5 (Orbital occupation budget). *Under (11) and (15), let a finite hidden register track a unit-vector command process as in (7). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(17) \quad \sum_{\substack{0 \leq t < N \\ \bar{K}_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048 D (16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (16) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas 3.2 and 3.3,

$$\mathbf{h}(f_{t+1}) - \mathbf{h}(f_t) \geq g_t \|\sqrt{f_t} - \Pi\sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma 3.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most $1 + D\tau^2$. The Gaussian maximum-entropy bound and (8) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (16). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary 3.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(18) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (9) with (17) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

4. UNIFORM GEOMETRY OF HERMITIAN CONJUGATION

Fix $q \geq 2$ and let $V = \text{Herm}_0(q)$ be the real space of traceless Hermitian q by q matrices, with inner product $\text{tr}(AB)$. Its dimension is $D = q^2 - 1$. Unitary conjugation is orthogonal on this space. The pure-state orbit has dimension $2(q-1)$, but normalized hidden centroids can have arbitrary spectra. We therefore prove a bound on *every* unit sphere orbit.

4.1. A Grassmann small-ball estimate. Let P_0 be a rank- k orthogonal projector in $\mathbb{C}^q$, $1 \leq k < q$, and let P be a Haar-distributed rank- k projector. Put $m = k(q-k)$ and

$$(19) \quad b_{q,k} = \frac{1}{m!} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)}.$$

Lemma 4.1 (Projector balls). *For $0 < R \leq 1$,*

$$(20) \quad \mathbb{P}(\|P - P_0\|_{\text{HS}} \leq R) \leq b_{q,k} R^{2k(q-k)}.$$

Proof. By invariance take P_0 onto the first k coordinates. Apart from a null set, a Haar subspace is the graph of $Z \in \mathbb{C}^{(q-k) \times k}$. Its density with respect to complex Euclidean Lebesgue measure is

$$(21) \quad \pi^{-m} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)} \det(I + Z^*Z)^{-q}.$$

For completeness, take a q by k matrix of independent standard complex Gaussian entries and split it into a square top block A and bottom block B . Its column space is Haar and $Z = BA^{-1}$ almost surely. The change $B = ZA$ has real Jacobian $|\det A|^{2(q-k)}$. Integrating A with weight $\exp[-\text{tr} A^*(I + Z^*Z)A]$ gives the determinant factor $\det(I + Z^*Z)^{-q}$. Complex Gaussian Gram-Schmidt gives $\mathbb{E}|\det A|^{2(q-k)} = \prod_{j=1}^k \Gamma(q-k+j)/\Gamma(j)$, since its squared orthogonal lengths have gamma shapes $1, \dots, k$. The powers of π give (21).

If θ_i are the principal angles, then $\|P - P_0\|_{\text{HS}}^2 = 2 \sum_i \sin^2 \theta_i$ and $\|Z\|_{\text{HS}}^2 = \sum_i \tan^2 \theta_i$. On the event in (20) each $\sin^2 \theta_i \leq R^2/2$, whence

$$\|Z\|_{\text{HS}}^2 \leq \frac{R^2}{2 - R^2} \leq R^2.$$

The density (21) is bounded by its value at zero. Multiply this value by the volume $\pi^m R^{2m}/m!$ of a real $2m$ -dimensional ball. This proves the assertion. $\square$

Define the explicit finite constant

$$(22) \quad A_q = \max_{1 \leq k < q} \left\{ \max(1, b_{q,k}) [9q(q-1)^2]^{k(q-k)} \right\}.$$

It is rational and at least one; no optimality is claimed.

Theorem 4.2 (All-spectra orbital anti-concentration). *For all $A, B \in \text{Herm}_0(q)$ of Hilbert-Schmidt norm one and all $0 < r \leq 1$,*

$$(23) \quad m_{\text{SU}(q)} \{U : \|UAU^* - B\|_{\text{HS}} \leq r\} \leq A_q r^{2(q-1)}.$$

In particular, the estimate is uniform at repeated eigenvalues and over mixed-state centroid directions. No lower bound on each eigenvalue gap is assumed.

Proof. Let $\lambda_1 \geq \dots \geq \lambda_q$ be the eigenvalues of A . Trace zero and $\sum_i \lambda_i^2 = 1$ imply $\lambda_1 - \lambda_q \geq q^{-1/2}$. Hence some k has

$$(24) \quad \lambda_k - \lambda_{k+1} \geq \delta_q := \frac{1}{\sqrt{q}(q-1)}.$$

For two conjugates C, D of A , let P_C, P_D be their top- k spectral projectors. Eigenvectors u_i of C and v_j of D obey

$$(\lambda_i - \lambda_j)\langle u_i, v_j \rangle = \langle u_i, (C - D)v_j \rangle.$$

Summing squares for $i \leq k < j$ and using equal projector ranks gives

$$(25) \quad \|P_C - P_D\|_{\text{HS}}^2 = 2 \sum_{i \leq k < j} |\langle u_i, v_j \rangle|^2 \leq 2\delta_q^{-2} \|C - D\|_{\text{HS}}^2.$$

If the event in (23) is empty there is nothing to prove. Otherwise choose one conjugate D in that event. Every other C in the event has $\|C - D\|_{\text{HS}} \leq 2r$. The image P_C is a Haar rank- k projector and lies within $2\sqrt{2}\delta_q^{-1}r$ of P_D . Set $c_q = 3\sqrt{q}(q-1)$, which is larger than this coefficient. If $c_q r \leq 1$, Lemma 4.1 gives a bound $b_{q,k}(c_q r)^{2k(q-k)}$. Since $k(q-k) \geq q-1$ and $r \leq 1$, this is at most the right side of (23). If $c_q r > 1$, then $A_q r^{2(q-1)} \geq c_q^{2(q-1)} r^{2(q-1)} > 1$, so the trivial probability bound suffices. Haar conjugation under $\text{SU}(q)$ and under $U(q)$ has the same law, completing the proof. $\square$

The proof isolates one uniformly separated spectral cut rather than requiring a simple spectrum. This is why pure-state and highly mixed centroids are handled by the same constant. The smallest possible nontrivial Grassmann exponent is $2(q-1)$.

Corollary 4.3 (Conjugation occupation). *For conjugation commands satisfying a group gap g_t , any hidden register with $q_N \geq \kappa > 0$ obeys*

$$(26) \quad \sum_{K_i \leq k} g_t \leq C_q \kappa^{-3-1/(q-1)} k^{1/(q-1)}, \quad C_q = 2048(q^2 - 1)(16A_q)^{1/(q-1)}.$$

For a constant gap $g > 0$ and wordwise error ε with $\kappa = \rho/\sqrt{q^2 - 1} - 2\varepsilon > 0$,

$$(27) \quad W_{N,\varepsilon} \geq \left(\frac{g\kappa^{3+1/(q-1)}}{C_q} N \right)^{q-1}.$$

Proof. Apply Lemma 3.1, Theorem 4.2 and Theorem 3.5, then use Lemma 2.2. $\square$

A group gap is stronger than a gap only on functions on the pure-state homogeneous quotient. The proof genuinely needs uniform control of the mixed-centroid directions, and does not silently replace this hypothesis by the weaker quotient gap.

5. SHARP MEMORY ON COMPLEX PROJECTIVE ORBITS

Write $r_q = \sqrt{1 - 1/q}$ and associate to a rank-one orthogonal projector P the unit vector

$$(28) \quad Y(P) = \frac{P - I/q}{r_q} \in \text{Herm}_0(q), \quad \mathcal{K}_q = \text{conv}\{Y(P) : \text{rank } P = 1\}.$$

The body $\mathcal{K}_q$ consists of normalized centered density matrices. It is invariant under conjugation and contains zero. Its extreme orbit is $\mathbb{CP}^{q-1}$. Fix an orthonormal real basis $F_1, \dots, F_D$ of $\text{Herm}_0(q)$ and use its coordinate queries in (5). One may take as seeds the q^2 rank-one projectors onto e_i , $(e_i + e_j)/\sqrt{2}$ and $(e_i + ie_j)/\sqrt{2}$, $i < j$. Their centered matrices affinely span $\text{Herm}_0(q)$. All seed and query alphabets are fixed when q is fixed.

5.1. A projective enclosure with quadratic loss. For rank-one projectors use the distance $d(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, the sine of their principal angle.

Lemma 5.1 (Finite projective synthesis). *For $0 < h \leq 1$, there is an h -net $\mathcal{V}_h$ of rank-one projectors with at most*

$$(29) \quad q(2\lceil 2\sqrt{2(q-1)/h} \rceil + 1)^{2(q-1)}$$

elements, and the entries of all these projectors can be chosen in $\mathbb{Q} + i\mathbb{Q}$. If $qh^2 < 1$, then

$$(30) \quad (1 - qh^2)\mathcal{K}_q \subseteq \text{conv}\{Y(P) : P \in \mathcal{V}_h\} \subseteq \mathcal{K}_q.$$

Proof. Every complex line has a representative with a largest coordinate scaled to one and the real and imaginary parts of all other coordinates in $[-1, 1]$. In each of the q charts round these $2(q-1)$ real coordinates to the grid of step $1/M$, where $M = \lceil 2\sqrt{2(q-1)/h} \rceil$. If v, w are the original and rounded representatives, their norms are at least one and $\|v - w\| \leq \sqrt{2(q-1)}/(2M)$. Normalization changes their distance by at most a factor two, and the sine distance is at most the distance of these normalized representatives. Thus the stated set is an h -net. The projector $w w^* / \|w\|^2$ has rational real and imaginary entries, and the grid count is (29).

For a nonzero traceless Hermitian H , let λ_1 and λ_q be its largest and smallest eigenvalues. A net projector within h of a top eigenline satisfies

$$\text{tr}(HP) \geq \lambda_1 - (\lambda_1 - \lambda_q)h^2 \geq (1 - qh^2)\lambda_1.$$

Indeed $-\lambda_q \leq (q-1)\lambda_1$ by trace zero. The support function of $\mathcal{K}_q$ at H is λ_1/r_q . The support-function criterion for convex containment proves (30). It applies to every direction, not just a finite set of tested queries. $\square$

Theorem 5.2 (A finite-alphabet exact upper machine). *For every finite unitary command alphabet and $0 < \rho \leq 1/10$, the experiment on (28) has an exact machine of width $O_q(N^{q-1})$. For each N , all command transition matrices can be chosen independent of the epoch. Each row uses at most q^2 successors. The final decoder may depend on N .*

Proof. Take $h = (8qN)^{-1/2}$ and $\eta = 1 - 1/(8N)$ in Lemma 5.1. Let V be the resulting finite set of centered vectors, and put $a_t = \eta^{-t}/4$. A label i represents $a_t Y(P_i)$ at epoch t . This is a proof description of a label; no matrix is stored in the register.

Since $\rho/a_0 = 4\rho \leq 2/5 < \eta$ and $0 \in \mathcal{K}_q$, any seed $\rho Y(P)$ is a convex combination of $a_0 V$. For each label i and each command U , the vector $\eta U Y(P_i) U^*$ lies in $\text{conv } V$. Choose a convex representation and use its weights as the stochastic transition row. The choices depend on the command and label but not on t . Multiplying by a_{t+1} shows that the expected represented vector is updated by exact conjugation, since $a_{t+1}\eta = a_t$.

Induction gives expected vector $\rho Y(U_w P U_w^*)$. The terminal column for query j is $a_N \text{tr}(F_j Y(P_i))$. It is legal because $|\text{tr}(F_j Y(P_i))| \leq 1$ and Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7$. Sampling a binary answer with this mean proves every conditional probability in (5) exactly. Carathéodory's theorem in real dimension $D = q^2 - 1$ permits q^2 successors. The number in (29) is $O_q(h^{-2(q-1)}) = O_q(N^{q-1})$. $\square$

The upper argument uses a quadratic support loss along the projective orbit. Applying a net on the full ambient sphere would instead give the larger exponent $(q^2 - 2)/2$.

5.2. Fixed finite algebraic alphabets. For $1 \leq j < q$ let $R_j, D_j \in \text{SU}(q)$ act as the identity outside the adjacent coordinate pair $(j, j+1)$ and on that pair as

$$(31) \quad R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} \zeta & 0 \\ 0 & \bar{\zeta} \end{pmatrix}, \quad \zeta = \frac{3 + 4i}{5}.$$

The alphabet has $4(q-1) + 1$ symbols: $\{I, R_j^{\pm 1}, D_j^{\pm 1} : 1 \leq j < q\}$. Assign probability $1/2$ to I and $1/[8(q-1)]$ to each other symbol.

Proposition 5.3 (Qualitative group expansion). *This command law has a norm gap on $L_0^2(\text{SU}(q))$ for every fixed q . Its entries and probabilities are algebraic. No numerical gap value is asserted for this family.*

Proof. The number ζ is not a root of unity: $\zeta + \zeta^{-1} = 6/5$ is rational but not an algebraic integer. The powers of R_j and D_j therefore have closures containing two distinct one-parameter subgroups in the corresponding embedded $\text{SU}(2)$. Their Lie brackets generate that $\mathfrak{su}(2)$. The adjacent embedded Lie algebras generate $\mathfrak{su}(q)$: adjacent elementary off-diagonal directions produce all other off-diagonal directions by brackets, and their self-brackets produce the traceless diagonal directions. The closed generated subgroup is consequently $\text{SU}(q)$.

The adjoint entries are algebraic. The spectral-gap theorem of Benoist–de Saxcé [2, Theorem 1.2] applies to an adapted algebraic measure on this compact connected group with simple Lie algebra. Equivalently, one may use the algebraic unitary result of Bourgain–Gamburd [3]. The symmetric average has an upper gap; weight $1/2$ at the identity makes its spectrum nonnegative. Its mean-zero operator norm is therefore strictly below one. This is the external full-group theorem, not a calculation in the defining q -dimensional representation. $\square$

Proof of Theorem 1.1. Corollary 4.3 gives the lower order, for every fixed positive calibration κ . Theorem 5.2 gives the matching exact upper order. Proposition 5.3 supplies fixed finite alphabets. The quantum assertion follows from the next proposition. Taking binary logarithms gives the label-bit formula. $\square$

Proposition 5.4 (A fixed quantum realization). *The projective experiment has an exact quantum realization on $\mathbb{C}^q$ using fixed seed resets, the prescribed unitary commands and fixed query measurements. It is independent of N .*

Proof. On seed P reset the density matrix to

$$\sigma_P = I/q + \rho Y(P) = (1 - \rho/r_q)I/q + (\rho/r_q)P.$$

This is positive and has trace one, since $0 < \rho \leq 1/10 < r_q$. Commands act by conjugation. For query j , use effects $(I + F_j)/2, (I - F_j)/2$. Hilbert–Schmidt normalization implies $\|F_j\|_{\text{op}} \leq 1$, so these are positive effects summing to I . Their outcome mean is $\text{tr}(F_j \sigma_{U_w P U_w^*}) = \rho \text{tr}(F_j Y(U_w P U_w^*))$. This is precisely (5). The seed reset is a channel, not a unitary; only one query is measured. $\square$

The comparison is a fixed q -dimensional quantum register versus $(q-1) \log_2 N + O(1)$ classical label bits. Dimension or entropy advantages for quantum models are prior phenomena [6, 17, 5]. The assertion here is the proved classical width order for this complete worst-word numerical behavior. It does not request jointly independent answers to all queries or optimize arbitrary adaptive experiments.

6. AN EFFECTIVE SIX-COMMAND RATIONAL EXAMPLE

The preceding group-gap application is qualitative. We now choose a separate arithmetic alphabet for which a classical theorem supplies a numerical infinite-dimensional norm bound. This does not evaluate the gap of the earlier five-command example.

Let R_x, R_y, R_z be rotations about the coordinate axes with cosine $-3/5$ and sine $4/5$; explicitly,

$$(32) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & 4/5 & -3/5 \end{pmatrix}, \quad R_y = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & -3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the fixed six-letter alphabet $\mathcal{A}_V = \{R_x^{\pm 1}, R_y^{\pm 1}, R_z^{\pm 1}\}$ and its uniform law. These are the adjoint rotations of the classical norm-five quaternions $1 \pm 2\mathbf{i}$, $1 \pm 2\mathbf{j}$, $1 \pm 2\mathbf{k}$, normalized by $\sqrt{5}$.

Proposition 6.1 (External arithmetic gap certificate). *For $P_V f = (1/6) \sum_{R \in \mathcal{A}_V} f(R^{-1} \cdot)$ on $L_0^2(\mathbb{S}^2)$,*

$$(33) \quad \|P_V\| \leq \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}.$$

Consequently $1 - \|P_V\|^2 \geq 4/9$.

Proof. This is the $p = 5$ specialization of the Lubotzky–Phillips–Sarnak spherical Hecke bound [10, 11]. In its unnormalized convention the $p + 1$ norm- p quaternion representatives act by a sum of rotations whose mean-zero norm is at most $2\sqrt{p}$. The representatives have positive odd real coordinate and even other coordinates when $p \equiv 1 \pmod{4}$; at $p = 5$ they are exactly the six displayed ones. The coauthor account [12, Section 3.2, the V -gate example] spells out these representatives and their Hecke interpretation. The unnormalized spherical norm convention is also stated in [13, Theorem 3(ii)]. Dividing by six gives (33). Passing from quaternions to adjoint rotations identifies $\mathbb{S}^2$ as the homogeneous quotient; inverse signs do not affect this symmetric average. The arithmetic bound is invoked as an external theorem, not reproved by finite harmonic tests. $\square$

Use six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, amplitude $\rho = 1/10$ and three final coordinate queries in (5).

Theorem 6.2 (Numerical finite-horizon width certificate). *For $0 \leq \varepsilon < 1/(20\sqrt{3})$ and every integer $N \geq 1$,*

$$(34) \quad \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N \leq W_{N,\varepsilon}(\mathcal{A}_V) \leq W_{N,0}(\mathcal{A}_V) < 150N.$$

The upper machine has exact rational command and decoder rows, common command matrices, and an N -dependent decoder. At zero error every internal selected cut has separate positive minimum four, while

$$(35) \quad W_{N,0} \geq \frac{N}{19\,906\,560\,000}.$$

The constants are proof constants, not sharp asymptotic constants.

Proof. For the standard $\mathrm{SO}(3)$ action the orbit of every unit vector is uniform on $\mathbb{S}^2$. A chordal cap of radius $r \leq 1$ has probability $r^2/4 \leq r^2$, so take $D = 3, s = 2, A = 1$ in Theorem 3.5. The sphere gap transfers to $L^2(\mathbb{R}^3)$ by polar coordinates. Thus $B_{3,1,2} = 98304$, and the squared gap $4/9$ gives the lower bound in (34). At $\varepsilon = 0$ the fourth power of $1/(10\sqrt{3})$ is $1/90000$, which yields (35).

Here is an explicit cardinality estimate for the upper construction. Take both rational stereographic charts

$$t \mapsto \frac{(2t_1, 2t_2, \pm(1 - \|t\|^2))}{1 + \|t\|^2}, \quad t \in [-1, 1]^2,$$

with a grid of step $1/M$ and $M = \lceil 2\sqrt{2N} \rceil$. Every hemisphere is covered by its chart on the unit disk. Each chart is 2-Lipschitz, so its grid images form a sphere net of chordal radius at most $\sqrt{2}/M \leq 1/(2\sqrt{N})$. Its convex hull contains η times the unit ball, with $\eta = 1 - 1/(8N)$, by the support function identity $u \cdot v = 1 - \|u - v\|^2/2$. There are at most $2(2M + 1)^2 \leq (82 + 48\sqrt{2})N < 150N$ labels. Use scales $a_t = \eta^{-t}/4$, initialization representing ρx , rows representing $\eta R u$ in this hull, and decoder $a_N e_j \cdot u$. The verification is the induction in Theorem 5.2. All data are rational; each nonempty rational convex system has rational weights, with at most four successors. The final decoder is bounded by $2/7$.

At a selected internal cut, fix any common command prefix. The six seed images affinely span $\mathbb{R}^3$, so the normalized continuation rank is four; future coordinate queries still separate these dimensions after any common suffix. For an upper factorization take a regular tetrahedron containing the radius- ρ ball and having circumradius $3\rho < 1$. Its four vectors give legal continuations under all future rotations. One embeds this selected-cut factorization in a complete machine by allowing uncapped raw histories elsewhere. Therefore the separate minimum is exactly four. This does not give a four-state machine at all cuts simultaneously. $\square$

For example, (35) exceeds four once $N \geq 79\,626\,240\,001$. The size of this number illustrates the conservative constants; the benefit is that no unspecified spectral constant is needed for this alphabet.

The associated qubit operations are $(I - 2i\sigma_j)/\sqrt{5}$ and their inverses, with seed Bloch vector $x/10$ and final Pauli query. They realize the numerical experiment exactly, independently of N . This quantum construction and the arithmetic gates are classical; (34) is the finite-register consequence. The earlier five-gate alphabet remains qualitatively gapped with a non-evaluated constant, and is retained unchanged in the supporting material.

7. OPERATIONAL AND THEOREM-LEVEL COMPARISONS

The comparisons below fix the minimized object before comparing results. They do not use a change of terminology as evidence of originality.

7.1. Ordered randomized programs.

Proposition 7.1 (The layered-program interpretation). *For a fixed command alphabet, Definition 2.1 is equivalent to a nonuniform probabilistic ordered read-once layered program with cut-dependent widths and real terminal columns. For one selected column, randomized Boolean acceptance has probability $(1 + m)/2$, where m is the column's mean, without changing any command-layer width. A separate charged terminal layer needs at most two labels. Maximum width is not total program size, and neither measure prices the row descriptions.*

Proof. At layer t , the outgoing edge weights labelled by command a are exactly the row $T_{t,a}(s, \cdot)$. Summing path probabilities gives (6). Conversely the edge weights of a probabilistic layered program form those stochastic matrices. At the terminal label s , toss a Bernoulli coin of parameter $(1 + d_j(s))/2$. This implements Boolean acceptance with the claimed probability. Choosing a query selects one column; it does not request independent outcomes for all columns. Padding a layer by unreachable labels and assigning normalized unused rows proves the same assertion for varying widths. $\square$

The equivalence permits arbitrary real edge probabilities and externally known layers. The Fourier-growth results for regular and width-three branching programs [15, 16, 9] have different structural assumptions and objectives. Here the external group average has a gap; the hidden transition matrices need not be regular or have a gap. In particular, a stochastic reset matrix is permitted. The all-spectra occupation inequality is a lower bound for the specified matrix-coefficient response, not a pseudorandom generator or a general lower bound for Boolean OBDDs.

7.2. Zero-delay coding and state aggregation. The structural coding results of Witsenhausen and Walrand–Varaiya, as formulated in [19, Theorems 1–2], reduce a Markov-source encoder to the current source and a posterior based on past transmitted messages. Wood–Linder–Yüksel prove stationary optimality and periodic finite-memory approximation for average distortion [19, Theorems 3–4]. A posterior is generally a point of a continuous simplex, not a finite alphabet. Those results optimize ongoing reproduction for a given stochastic source. Here a machine sees commands rather than the true orbit vector, and every command word must satisfy a final

numerical specification. Only its current label, not a free transcript of sent messages, is retained. Thus a sufficient belief state in the coding problem does not give a cardinality upper bound in the present problem. Conversely our width lower bound does not lower-bound the optimal average distortion of those coders.

Nonanticipative rate-distortion minimizes directed information over causal reproduction kernels under expected distortion constraints. For example, the finite-horizon Markov-source analysis in He–Charalambous–Stavrou [7, Sections II–III] derives a dynamic optimization of those kernels and an alternating computation. That objective is neither $\max_t K_t$ nor $\sum_t K_t$. In particular, mutual information can be small for a channel whose exact finite positive realization needs many labels. A source-law rate-distortion optimization cannot be inserted in place of the wordwise specification without a separate reduction.

Geiger–Tommel [4, Theorem 9] characterize higher-order strong lumpability by an equality of conditional entropies for a stationary finite Markov chain and a deterministic lumping. Their entropy-rate preservation results concern the whole lumped process. We allow randomized, epoch-dependent maps, retain a command-driven terminal observable, and do not require preservation of the trajectory entropy rate or of a Markov partition. State aggregation is therefore an adjacent comparison, not a theorem claimed to have been subsumed. The common use of conditional expectations does not identify the optimization tasks.

7.3. Conditional expectations and Gaussian entropy. Atalik–Köse–Gastpar [1, Theorem 1 and Proposition 2] study $M = \mathbb{E}[X \mid X + W]$ for Gaussian observation noise. In the identity observation case their inequality has the form $h(M) \geq 2h(X) - h(X + W)$, with a vector extension. Lemma 3.2 instead permits an arbitrary finite conditioning variable J , adds new independent noise *after* conditioning, and upper-bounds a difference of two smoothed entropies by the martingale coupling’s squared displacement. A discrete M is allowed here, whereas its unsmoothed differential entropy is not the quantity being compared. The two displayed inequalities are not direct substitutions of one another. The Gaussian score and posterior-covariance identities used in (13) are classical, and our short semiconcavity proof is not claimed as a new estimation identity. Polyanskiy–Wu’s Wasserstein entropy continuity [14] provides a further comparison; the quadratic cost here uses the particular conditional-mean coupling, not arbitrary transport. The argument needed for the occupation theorem is the amortization of that cost together with orbital sparsity.

7.4. Positive realization and orbit geometry. Positive realization and invariant convex sets have a long history [8, 18]. The occupation theorem does not require an internal state to be a point on an observed orbit or in the residual span. Such a restriction would miss hidden stochastic lifts. Classical sphere and projective covering exponents explain the net counts; the additional issue for conjugation is uniformity over spectra of all conditional centroids. Theorem 4.2 settles that issue through a separated spectral projector and the explicit Grassmann bound. Its exponent is matched by an exact projective, rather than ambient spherical, construction. This is the reason that $q - 1$, not $(q^2 - 2)/2$, appears in Theorem 1.1.

The deep group gaps are external inputs. Proposition 5.3 uses a qualitative theorem; Proposition 6.1 uses a numerical Hecke norm theorem for a different named alphabet. Finite diagonalization of a few harmonics is not a proof of either full gap. Neither density of a subgroup nor a gap in its defining finite-dimensional representation is silently promoted to a full L^2 norm gap.

7.5. Implementation and the remaining distinctions. All exact theorems here count atomic nonuniform labels. Replacing each row by a rational approximation and then using a fixed-duration padded fair-bit comparison sampler, as in the retained predecessor, incurs additional control labels and terminal approximation error. The padding removes dependence of duration on private coin outcomes; it does not price the read-only threshold tables or prove uniform space complexity. No exact finite-coin compiler for arbitrary irrational rows is asserted.

The proved results are the representation-valued occupation inequality, the all-spectra conjugation estimate, the matched projective cardinality order, and the effective six-command consequence. They do not classify all compact representations, all finite alphabets, or arbitrary adaptive experimental design. The sufficient calibration threshold is not shown optimal, and signal approaching zero with the horizon requires retaining the explicit constants rather than invoking a fixed-signal asymptotic. Exact finite integer optima and leading constants remain different questions. The standalone article contains all proofs used for its new claims; preservation records and computation are not mathematical or priority certificates.

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